# CSE 311 - Data Communication

## Error Correcting Code

code টা এমন ভাবে বানাবো যেন error auto correct করা যায়।

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# Outline

- Linear Block Codes
- Cyclic Codes
- Convolutional Codes
- Viterbi Decoding

# Linear Block Codes

$$\boldsymbol{c} = (c_1, c_2, \ldots, c_n)$$ codeword

$$\boldsymbol{d} = (d_1, d_2, \ldots, d_k)$$ actual data

$\underline{n > k}$

$n-k$ সংখ্যক check bit আছে

$$\boxed{\boldsymbol{c} = \boldsymbol{d}\boldsymbol{G}}$$

$G \rightarrow$ generator matrix

actual data $d = 101$

$\underbrace{101}_{\text{data}}\ \underbrace{00}_{\text{error correcting code}}$ $(dG)$

$$\boldsymbol{G} = \left[\begin{array}{ccccc|cccc} 1 & 0 & 0 & \cdots & 0 & h_{11} & h_{21} & \cdots & h_{m1} \\ 0 & 1 & 0 & \cdots & 0 & h_{12} & h_{22} & \cdots & h_{m2} \\ & & & \vdots & 0 & & & \vdots & 0 \\ 0 & 0 & 0 & \cdots & 1 & h_{1k} & h_{2k} & \cdots & h_{mk} \end{array}\right]$$

$\underbrace{\boldsymbol{I}_k(k \times k)}_{\text{identity}}$ $\underbrace{\boldsymbol{P}(k \times m)}_{\text{কিছু coefficients}}$

$G = [I_k \quad P]$

$dG = [dI_k \quad \underbrace{dP}_{\text{check bit}}]$

multiply করা হলে first part এ data ই থাকবে।

$$\begin{aligned} \boldsymbol{c} &= \boldsymbol{d}\boldsymbol{G} \\ &= \boldsymbol{d}[\boldsymbol{I}_k \quad \boldsymbol{P}] \\ &= [\boldsymbol{d} \quad \boldsymbol{d}\boldsymbol{P}] \\ &= [\boldsymbol{d} \quad \boldsymbol{c}_p] \end{aligned}$$

$$\boxed{\boldsymbol{c}_p = \boldsymbol{d}\boldsymbol{P}}$$

# Linear Block Codes

A block code is a **linear block code** if for every pair of codewords $\mathbf{c}_a$ and $\mathbf{c}_b$ from the block code,

$$\mathbf{c}_a \oplus \mathbf{c}_b$$

is also a codeword. For this reason, linear codes must have an all-zero codeword $\mathbf{000 \cdots 00}$. For linear codes, the **minimum distance** equals the **minimum weight**.

XOR → point wise multiplication

# Decoding

Let us consider some codeword properties that could be used for the purpose of decoding. From Eq. (14.8) and the fact that the modulo-2 sum of any sequence with itself is zero, we get

$$\mathbf{d} \cdot \mathbf{P} \oplus \mathbf{c}_p = \underbrace{[\mathbf{d} \quad \mathbf{c}_p]}_{\mathbf{c}} \begin{bmatrix} \mathbf{P} \\ \mathbf{I}_m \end{bmatrix} = 0 \tag{14.9}$$

$c_p \oplus c_p = 0$

where $\mathbf{I}_m$ is the identity matrix of order $m \times m$ $(m = n - k)$. Thus,

$$\mathbf{c}\mathbf{H}^T = 0 \tag{14.10a}$$

where

$$\mathbf{H}^T = \begin{bmatrix} \mathbf{P} \\ \mathbf{I}_m \end{bmatrix} \tag{14.10b}$$

and its transpose

$$\mathbf{H} = [\mathbf{P}^T \quad \mathbf{I}_m] \tag{14.10c}$$

$$\boldsymbol{r} = \boldsymbol{c} \oplus \boldsymbol{e}$$

s থেকে detect করতে পারি error হয়েছে কিনা

$$\boldsymbol{s} = \boldsymbol{r}\boldsymbol{H}^T$$

r → received signal

problem:

1bit/2bit error

এর জন্য s same হতে পারে। তখন check করি কোন error এর probability highest।

$$= (\boldsymbol{c}_i \oplus \boldsymbol{e}_i)\boldsymbol{H}^T$$

$$= \boldsymbol{c}_i H^T \oplus \boldsymbol{e}_i \boldsymbol{H}^T$$

$$= \boldsymbol{e}_i \boldsymbol{H}^T$$

original code টা চলে যায়। শুধু error টা পাই।

$$P(\boldsymbol{r}|\boldsymbol{c}_i) = P_e^d (1 - P_e)^{n-d} = (1 - P_e)^n \left(\frac{P_e}{1 - P_e}\right)^d$$

যেটার probability high সেটাকে নিবো

# Cyclic Codes

property : shift করলেও আরেকটা code পাওয়া যায়। Polynomial আকারে লেখা হয়।

left shift 1001 → 0011

$$c = c_1 x^{n-1} + c_2 x^{n-2} + \dots + c_n x^0$$

$$c = x^3 + 0 + 0 + 1$$ (as 4 bit এর)

example:

$$d = x^3 + x$$ (1010)

$$g = x^3 + x^2 + 1$$

generator polynomial

$$c = d \cdot g = \qquad \rightarrow \frac{\qquad\qquad}{x^n + 1}$$

$x^7$ এর বড় term বাদ দেয়া হয়।

(codeword এর থেকে বড়)

$x^6$